Effective rate, physical green service and descriptor births

8 October 2026 — continuation 67; qualification-scope gate Q.ACT

Scope. Local analytical diagnostics. They do not solve the joined traffic model, exhibit reachable common-control scenarios, prove complete TST benefit or certify gradients. Traffic mass remains continuous. No alternative actuator or birth law is adopted.

1. Averaging permission and limiting receiving need not commute

Let $s > 0$ be fully open entrance service (vehicle equivalents/s), $R \geq 0$ constant receiving (same units), and $\eta \in [0, 1]$ a fixed green fraction. Assume enough waiting mass to saturate entrance demand throughout this diagnostic. Let $b(t) \in \{0, 1\}$ be a periodic signal with cycle $c > 0$ and green duration ηc. Its nominal rate is $r = \eta s$. The current ideal cap gives

$$q_A = \min\{\eta s, R\}. \tag{1}$$

With instantaneous first-order release and no startup loss, the pulse gives

$$q_P(t) = \min\{sb(t), R\}, \qquad \overline{q}_P = \frac{1}{c}\int_0^c q_P(t)\,dt = \eta \min\{s, R\}. \tag{2}$$

Hence the discrepancy is

$$q_A - \overline{q}_P = \begin{cases} (1-\eta)R, & 0 \leq R \leq \eta s, \\ \eta(s-R), & \eta s < R < s, \\ 0, & R \geq s. \end{cases} \tag{3}$$

It is positive for $0 < \eta < 1$ and $0 < R < s$, with maximum

$$\max_{R \geq 0}(q_A - \overline{q}_P) = \eta(1-\eta)s \quad \text{at } R = \eta s. \tag{4}$$

For $s = 1$, $R = 0.4$ and $\eta = 0.5$, $q_A = 0.4$ but $\overline{q}_P = 0.2$ vehicle equivalents/s. Over one diagnostic ten-second cycle, they admit four and two equivalents respectively. These are uncalibrated local inputs, not a traffic simulation. The averaged discrepancy is independent of c; decreasing cycle length does not repair the wrong operator.

With finite bin duration Δt, available mass B and receiving mass R, M.A is

$$A_A = \min\{B, r\Delta t, \mathsf{R}\}. \tag{5}$$

Pair its debit/credit: $W^+ = W + I - A_A$, $n^+_{\text{app}} = n_{\text{app}} + A_A$ before other transfers. Conservativity does not prove pulse equivalence.

2. What the inspected continuum-signal source actually averages

For the controlled incoming road in Han et al.'s author preprint, $D(t)$ is demand (veh/s), C its capacity, and $S(t)$ target supply. Its equations 2.7–2.8 distinguish

$$q_{\text{on/off}}(t) = \min\{D(t), b(t)\min(C, S(t))\}, \qquad q_{\text{cont}}(t) = \min\{D(t), \eta\min(C, S(t))\}. \tag{6}$$

This is not $\min\{D(t), \eta C, S(t)\}$. Its continuum signal scales effective receiving, not merely nominal capacity before receiving. The preceding section's saturated example matches the distinction. This is source evidence for candidate B's family, not a verified full ramp adaptation.

Under the source's no-spillback hypotheses, its Theorem 4.4 bounds the on/off–continuum cumulative count difference by

$$\left|N^c(t,x) - N^0(t,x)\right| \le \eta(1-\eta)c\min(C_1, C_3). \tag{7}$$

Both counts use that source's link, boundary and signal conditions. Its spillback sections give distinct limitations; its first-order signal model does not include our endogenous bounded-acceleration merge descriptors. This bound therefore proves neither M.A equivalence nor descriptor/gradient equivalence. No error tolerance or guaranteed study regime is inferred.

3. The deterministic birth remainder is independent of time refinement

For a directed generating stream, let $H(t)$ be cumulative actual transferred mass, $m_* > 0$ the fixed descriptor threshold, and $K(t_k)$ the cumulative generated count at sample t_k (with declared public initial progress). For zero initial progress,

$$K(t_k) = \lfloor H(t_k)/m_* \rfloor, \qquad e(t_k) = H(t_k) - m_* K(t_k), \quad 0 \le e(t_k) < m_*. \tag{8}$$

At a fixed terminal $H(T)$, refining the sampled time mesh leaves $\lfloor H(T)/m_* \rfloor$ and $e(T)$ unchanged. At $H(T) = 0.4m_*$, no record is generated. The transferred $0.4m_*$ is still in the field, exits and complete inventory accounting. No physical fractional vehicle is asserted to lack a real blockage.

4. A small count error is not automatically a small event error

For $0 < \epsilon < m_*/2$, a local threshold diagnostic gives

$$H_- = m_* - \epsilon, \quad H_+ = m_* + \epsilon, \qquad \lfloor H_-/m_* \rfloor = 0, \quad \lfloor H_+/m_* \rfloor = 1. \tag{9}$$

The difference 2ϵ can be arbitrarily small while the record count differs. This disproves an unconditional inference from small flow/count error to identical descriptor sets. It supplies no lower bound on TST error: a generated record may be nonbinding or its effect may be small.

For a finite declared sample set, one possible conditional statement is

$$\max_k |\widetilde{H}(t_k) - H(t_k)| < \gamma, \qquad \min_k \operatorname{dist}(H(t_k), m_*\mathbb{Z}) > \gamma \implies \lfloor \widetilde{H}(t_k)/m_* \rfloor = \lfloor H(t_k)/m_* \rfloor. \tag{10}$$

Here $\gamma > 0$ is a hypothetical mass margin, not an adopted tolerance or an exclusion of threshold cases. Equal sampled counts still do not imply equal speeds, effects, intermediate timing or derivatives.

For completeness, the source's reported Poisson alternative at mean increment $\lambda = \Delta H/m_*$ has

$$\Pr(B = 0) = e^{-\lambda} > 0, \qquad \mathbb{E}[B] = \lambda. \tag{11}$$

It does not guarantee a record for every positive increment. No randomness, new seed contract or control-dependent event differentiation is adopted.

5. Scope of the gate

Candidate A retains M.A's effective-rate operator and the selected source birth/reset as an explicitly idealized reference to qualify further. Candidate B changes the operator to a source-supported supply-scaled continuum signal family. Candidate C qualifies finite pulses and launch response. The decision is scientific scope/actuation, not storage or computation cost.

Exact retention applies after generation. None of these local identities closes the own-speed/traffic/node feedback or clearance. A useful physical gradient remains a separate research claim.

Evidence: supplied Laval–Daganzo journal, Section 2.2/Appendix A; Han et al., author preprint arXiv:1309.3461v2, equations 2.7–2.8, Theorems 4.1/4.4 and Sections 5–6; FHWA Handbook Section 5.3.6. Source versions, access boundaries and screening are in the linked Markdown audit.